

NightWalk : Predicting Nighttime Luminesce from Daytime SVI

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Abstract

We propose NightWalk, a computer vision system that predicts relative nighttime street brightness directly from daytime imagery. While previous methods rely on sparse nighttime data or low-resolution satellite imagery, our approach leverages daytime visual cues — such as street lamps, storefronts, and tree canopy — to estimate on-the-ground lighting conditions. We train a two-stage model using weakly supervised labels from Grounding DINO and fine-tune it on a novel dataset of 974 manually paired day–night images from New York City. Our results demonstrate that daytime imagery contains a genuine signal for nighttime illumination that is recoverable by deep vision models but cannot be reduced to simple object counts: while a counting-based baseline fails to capture meaningful variation ($R^2 = 0.081$), our fine-tuned model achieves $R^2 = 0.417$. We further demonstrate the utility of these predictions through a prototype lighting-aware route planner that offers safer, well-lit pedestrian paths. Our code is released in [this repository](#).

1. Introduction

Street lighting influences perceptions of safety and the routes pedestrians choose at night [1, 8], yet existing navigation tools like Google Maps do not account for lighting conditions.

A solution would be to collect nighttime street-level imagery to map urban illumination. However, nighttime street view imagery (SVI) is scarce: collecting it requires specialized equipment and conditions [3], and providers such as Google Street View cover far fewer locations at night than during the day [3]. Satellite imagery, while more consistently available at night, captures lighting from above rather than at the pedestrian level, making it a poor proxy for the on-the-ground experience of walking through a street [3].

We propose NightWalk, a computer vision system that works within current data constraints by predicting nighttime street brightness directly from daytime imagery. Our approach is motivated by the observation that daytime images contain visual cues (*i.e.* street lamps, commercial

storefronts, and tree canopy) that may be predictive of how bright a street will be at night. We first train a model to detect these lighting-relevant features from daytime images using weakly supervised labels derived from Grounding DINO [4], then attach a brightness regression head and fine-tune it on a novel dataset of 974 manually collected day–night image pairs. We conduct a controlled backbone ablation study to evaluate whether task-specific pretraining on urban visual features improves over standard initialization. We find that while our counting-based pretraining does not surpass ImageNet initialization, deep vision models fine-tuned on our paired data recover meaningful nighttime illumination signal from daytime imagery alone, but this signal cannot be reduced to simple object counts. At inference, NightWalk takes only a daytime image as input and predicts the expected nighttime brightness of the depicted scene. We demonstrate a downstream application to well-lit pedestrian route planning.

Our contributions are as follows:

- A two-stage model that predicts nighttime street brightness from daytime imagery, requiring only a small set of paired day–night images for supervision.
- A weakly supervised labeling pipeline using Grounding DINO to automatically annotate lighting-relevant features in daytime street-view imagery.
- A dataset of 974 paired day–night street-view images collected across New York City.
- A downstream application to pedestrian route planning using predicted brightness estimates.

2. Related Work

Assessing pedestrian nighttime lighting has been approached through satellite imagery, ambient sensors, and street view imagery (SVI), each facing limitations in scalability or data availability.

Satellite-based approaches capture aggregate urban illumination but lack the street-level resolution needed to assess pedestrian lighting [6]. Sensor-based methods like NightLight [1] achieve accurate sidewalk-level measurements via smartphone light sensors. A user study found that all participants changed their routes when provided with lighting data, but the approach requires active data collection along



Figure 1. Overview of the NightWalk pipeline.

every path.

SVI offers a ground-level perspective unavailable from satellite imagery, though approximately 95% of studies rely on daytime imagery [5]. Most relevant to our work, Fan et al. [3] train a model on 2,831 day-night image pairs to predict nighttime brightness, validating against luminance meters and scaling to 50,689 images across Singapore to produce a city-wide illumination map.

3. Methods

Our proposed approach has two stages (Figure 1): First, we train a model to detect lighting-relevant visual features (*i.e.* street lamps, commercial storefronts, and tree canopy) from daytime images, using weakly supervised labels derived from Grounding Dino [4]. Then, we add brightness regression head to this model and fine-tune it using a set of 1,000 day-night image pairs, where the nighttime image provides the brightness supervision signal in the form of pixel-brightness. At inference, the model takes only a daytime image as input and predicts how bright that street will be at night.

3.1. Nighttime SVI Collection

Since our original dataset does not guarantee co-located daytime and nighttime imagery, we collected our own nighttime SVI by visiting the same locations and orientations as a subset of the daytime photos. A map of the locations of our collected images can be found in the Appendix 8b. Images were captured on foot in New York City, with the photographer positioned at approximately the same vantage point as the original fleet vehicle.

To match nighttime images to their daytime counterparts, we developed a semi-automatic annotation tool that accepts the GPS coordinates and heading of a captured nighttime photo, retrieves the top- N spatially proximate images from the daytime dataset, and allows a human annotator to confirm the correct match, saving the pair to a CSV file. A screenshot of our tool can be found in the Appendix (Figure 9). Manual supervision was necessary because the GPS coordinates recorded by the fleet vehicles were not always consistent with those recorded by our phones, making fully automatic matching unreliable. Using this process, we iden-

tified 974 nighttime-daytime pairs, which formed our final training set.

3.2. Pixel Brightness Computation

Ground truth brightness is computed from nighttime images by converting to grayscale and computing the mean pixel brightness intensity, and then computing the z-score normalizing across the dataset.

$$\text{gray_mean_zscore} = \frac{\mu_i - \bar{\mu}}{\sigma} \quad (1)$$

where μ_i is the mean grayscale intensity of image i , and $\bar{\mu}$, σ are the dataset mean and standard deviation. Instead of predicting absolute pixel values, we train our model to predict how bright a street is relative to the average street in our dataset. This is fitting for our application, as a pedestrian chooses between routes relative to other candidate routes' brightness.

3.3. Lighting Feature Labeling via Grounding DINO

An assumption we make in training NightWalk is that three daytime-visible features can serve to predict nighttime brightness: street lamp density, commercial storefront presence, and tree coverage. To automatically label these features at scale, we use Grounding DINO [4], an open-vocabulary detector that accepts free-form text prompts, to detect relevant objects in each daytime image. Specifically, we query the model with the prompt "*tree . tree top . tree trunk . streetlight . street lamp . light pole . shop front . storefront .*" where period-separated phrases.

We then map raw detection labels to our three semantic categories via regular expression matching: tree-related terms (e.g. *tree*, *tree trunk*) aggregate into tree canopy counts, lamp-related terms (e.g., *street lamp*, *light pole*) into street lamp counts, and facade-related terms (e.g., *storefront*, *shop front*) into commercial storefront counts. For all three features, we use raw detection counts rather than binary presence/absence labels. For each detection, Grounding DINO also returns a bounding box, which we use to compute the total pixel area occupied by each category. These area sums capture object prominence in the scene and are used as additional supervision targets during backbone pretraining.

3.4. Feature Detection Backbone Model

To obtain scene-aware backbone weights, we supervise EfficientNet-B0 [7] on daytime images using object counts produced by Grounding DINO [4]. We adopt EfficientNet-B0 initialized on ImageNet following [3], and replace the classification head with one independent regression head per target, each consisting of two fully-connected layers (1280→128→1) with ReLU activations and dropout ($p =$

0.5, $p = 0.3$), and a final ReLU to ensure non-negative predictions. All heads share the same backbone.

We train two variants. **DINO-counts v1** uses three targets—raw detection counts for trees, streetlights, and storefronts. **DINO-counts v2** extends v1 with three bounding-box area targets per category, giving six heads total, to capture the visual prominence of each feature type rather than presence alone. The resulting backbone weights are then used to initialize the brightness regression model (Section 3.5), replacing standard ImageNet initialization with representations tuned to our daytime street data.

Since our dataset consists of images from cars that made repeated passes along the same streets, naive random sampling would yield a corpus dominated by near-identical viewpoints, as a single vehicle may traverse the same block dozens of times. To mitigate this, we apply a stratified sampling procedure before running the zero-shot detector and training EfficientNet-B0. The sampled image coverage can be found in the appendix in Figure 8a.

We first discretize each image’s map-matched GPS coordinates into a spatial grid of approximately 50-meter cells (binning snapped latitude and longitude at a resolution of 0.0005°). We then group images by the combination of grid cell and vehicle identifier, retaining at most three images per group. This ensures that no single vehicle dominates any given location while still preserving multiple viewpoints per cell—capturing variation in lighting, occlusion, and pedestrian presence across repeated passes. This procedure yields approximately 13,000 candidate images.

Finally, we enforce a strict partition between the EfficientNet pretraining corpus and the day-night pairs used for subsequent fine-tuning. All daytime images appearing in the paired fine-tuning set are excluded from the pretraining pool prior to sampling, and we verify that the two sets are disjoint before running experiments.

3.5. Brightness Regression Head

To predict nighttime brightness, we discard the pretraining heads and attach a new regression head to the frozen backbone: two fully-connected layers (1280→256→1, ReLU, dropout $p = 0.3$), producing a single scalar prediction of `gray_mean_zscore` (1). The entire network is then fine-tuned end-to-end on our 974 day–night pairs using using Huber loss.

We evaluate using 5-fold cross-validation grouped by day image, so that all night photos matched to the same daytime scene appear in the same fold. This prevents the model from implicitly memorizing location-specific appearance during training and evaluating on a nearby viewpoint of the same scene. We evaluate three backbone initializations—ImageNet, self-supervised SimCLR pretraining (SSL), and DINO-counts (Section 3.4)—keeping all other hyperparameters fixed to isolate the effect of pretraining.

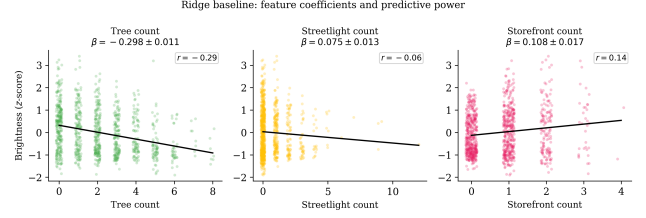


Figure 2. Ridge regression baseline: per-feature coefficients and their relationship to nighttime brightness. Each panel shows a DINO object count vs. brightness (z-score) with fitted slope β and Spearman r .

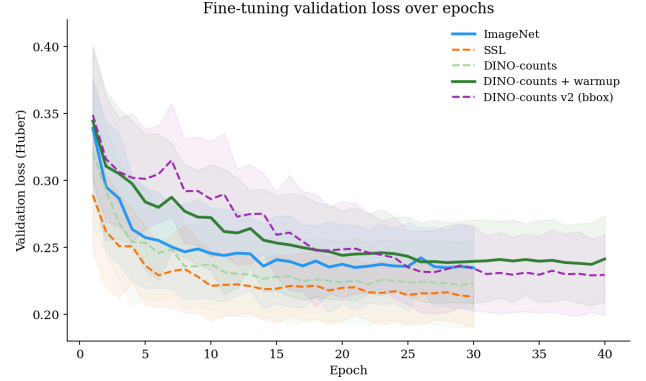


Figure 3. Validation loss curves during fine-tuning for each backbone initialization (mean $\pm$ std across 5 folds).

3.6. Training Details

Pretraining – We train using AdamW ($\text{lr} = 1 \times 10^{-4}$, weight decay 10^{-3}) with cosine annealing over 100 epochs, batch size 64, and Huber loss. Gradients are clipped to ℓ_2 norm ≤ 1.0 .

Fine-tuning – We use AdamW with batch size 16 over 40 epochs and Huber loss. We apply a warmup to our EfficientNet backbone, freezing it for the first 10 epochs to allow the regression head to stabilise, then unfreeze with differential learning rates (backbone $\text{lr} = 1 \times 10^{-5}$, head $\text{lr} = 3 \times 10^{-4}$). For ImageNet and SSL backbones, a uniform $\text{lr} = 3 \times 10^{-5}$ is used with no warmup.

4. Results and Analysis

We evaluate NightWalk across two experimental conditions. First, we establish a counting-based linear baseline to assess whether DINO-derived object counts are predictive of nighttime brightness on their own. Second, we conduct a backbone ablation to determine whether task-specific pretraining improves over standard ImageNet initialization when fine-tuning on paired day–night images.

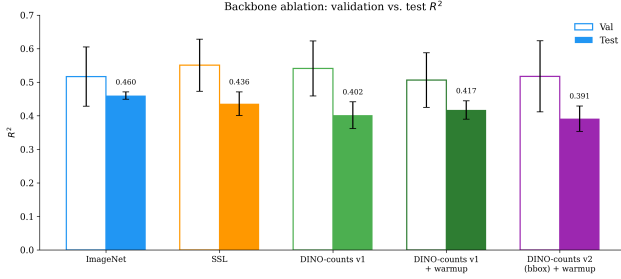


Figure 4. Test R^2 by backbone initialization (mean $\pm$ std across 5 folds). ImageNet achieves the highest test performance despite DINO-counts backbones reaching similar validation loss.

4.1. Counting-Based Linear Baseline

As a baseline, we fit a ridge regression from DINO counts directly to `gray_mean_zscore` (1) using 5-fold cross-validation on our 974 paired images, yielding $R^2 = 0.081$. Figure 2 shows the per-feature coefficients and their relationship to brightness. Tree count is negatively associated with brightness, consistent with tree canopy blocking artificial light sources at night. Streetlight and storefront counts carry positive but weak coefficients ($\beta = 0.075$ and 0.108 respectively), suggesting these features provide limited discriminative signal at the count level. The low overall R^2 indicates that object counts alone are insufficient for brightness prediction, motivating the use of an end-to-end visual model capable of extracting richer spatial and contextual patterns from the full daytime image.

4.2. Backbone Ablation

To assess whether our feature-counting pretraining improves brightness prediction, we compare five backbone initializations: ImageNet (standard supervised pretraining), SSL (SimCLR [2] pretrained on our 13k daytime images without labels), and three DINO-counts variants — v1 (raw object counts), v1 with warmup scheduling, and v2 with additional bbox-area supervision. All other hyperparameters are held fixed to isolate the effect of pretraining.

Figure 3 shows validation loss curves during fine-tuning. All backbones converge to similar validation loss by epoch 40, suggesting that fine-tuning successfully adapts each initialization to the brightness regression task. However, DINO-counts variants exhibit slightly higher variance across folds, indicating less stable convergence than the ImageNet and SSL baselines.

Table 1 and Figure 4 report test-set performance. Despite similar validation loss, DINO-counts variants generalize worse than ImageNet, with a consistent val-test gap that warmup scheduling and bbox supervision do not close. The best DINO-counts variant (v1 + warmup, $R^2 = 0.417$) trails ImageNet ($R^2 = 0.460$) by 0.043, while v2 with bbox supervision performs no better ($R^2 = 0.399$). SSL pretrain-

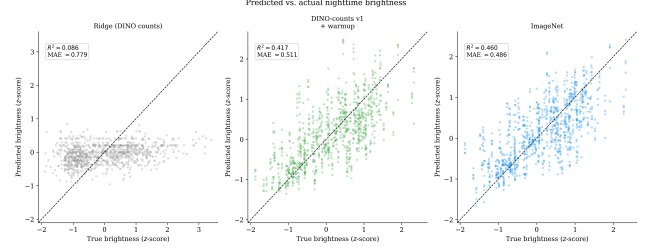


Figure 5. Predicted vs. actual nighttime brightness for the ridge baseline with DINO counts (grey), DINO-counts backbone (green), and ImageNet backbone (blue). Each point is one test image; the dashed line indicates perfect prediction.

ing on urban imagery achieves $R^2 = 0.436$, close to ImageNet, suggesting that self-supervised features learned from domain-relevant scenes transfer nearly as well as supervised ImageNet representations without requiring labeled data. Together, these results suggest the task mismatch between object counting and brightness prediction is fundamental: features tuned for detecting and sizing objects do not transfer as well as either general ImageNet or self-supervised urban representations.

4.3. Final Model Results

Figure 5 shows predicted vs. actual brightness for the ridge baseline, DINO-counts v1 with warmup (ours), and the ImageNet backbone (best-performing). The ridge baseline clusters near the dataset mean regardless of true brightness, confirming that raw object counts are insufficient for this task. DINO-counts v1 with warmup shows a clearer positive trend ($R^2 = 0.417$, MAE = 0.511) but with high variance, reflecting its less stable convergence across folds. The ImageNet model produces the tightest spread around the diagonal ($R^2 = 0.460$, MAE = 0.486), capturing meaningful variation in nighttime brightness from daytime imagery alone. These results suggest that daytime street-view images contain genuine signal about nighttime illumination that are recoverable by a deep vision model, but cannot be reduced to simple object counts (at least for the objects we identified).

4.4. Qualitative Results

To qualitatively analyze the final model, we inspect our day-night pairs and their predicted normalized nighttime brightness scores, `gray_mean_zscore`.

Figure 6 shows sample outputs from the brightness model. Each column contains the daytime image used as input, the matched nighttime image used as the target, and the model’s predicted brightness score. The resulting pairs make it possible to visually inspect whether daytime features actually correspond to nighttime brightness. The images span dark, medium, and bright nighttime conditions.

Table 1. Test-set brightness prediction performance (5-fold CV, grouped by scene). Mean $\pm$ std across folds. $\uparrow$ higher is better, $\downarrow$ lower is better.

Model	Backbone init.	MAE $\downarrow$	RMSE $\downarrow$	R^2 $\uparrow$
Ridge regression	DINO counts	0.779	0.955	0.081
EfficientNet-B0	ImageNet	0.486 ± 0.009	0.634 ± 0.006	0.460 ± 0.011
	SSL	0.498 ± 0.017	0.648 ± 0.020	0.436 ± 0.035
	DINO-counts v1	0.509 ± 0.016	0.667 ± 0.022	0.402 ± 0.040
	DINO-counts v1 + warmup	0.501 ± 0.019	0.655 ± 0.020	0.417 ± 0.025
	DINO-counts v2 + warmup	0.513 ± 0.019	0.669 ± 0.018	0.399 ± 0.032

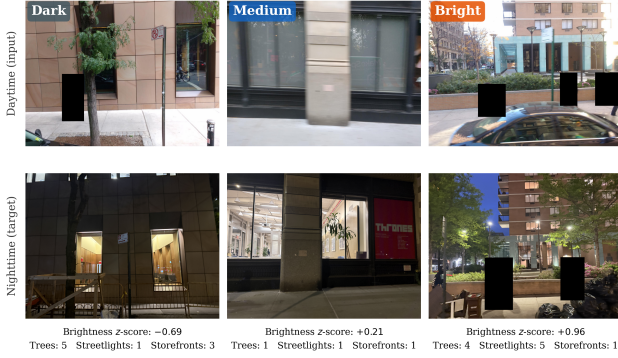


Figure 6. Qualitative examples of NightWalk brightness predictions. Each column shows a daytime input image, the matched nighttime image, and the model’s predicted brightness score. These examples demonstrate how the model outputs a predicted scalar brightness that broadly separates into dark, medium, and bright scenes.

In the darker example of Figure 6, the nighttime scene contains limited visible illumination and receives a low predicted brightness score. In the brighter example, the night image includes stronger visible lighting from storefronts and street-level illumination, corresponding to a higher predicted score. This qualitative pattern supports the quantitative results in Section 4.3, suggesting that the model learns visual features from the daytime images that are, at least in some cases, predictive of nighttime illumination.

However, the qualitative examples also highlight several sources of noise in the pairing process. Even when images are taken from the same street segment, the iPhone photo may differ from the original daytime image in camera height, angle, and framing. These differences can affect the apparent brightness of the nighttime scene and introduce error into the model’s supervision signal. Additionally, the daytime images were taken 10 years prior to the nighttime images. Thus, many of the structures in the scenes have changed during that time. As a result, the day-night pairs should be interpreted not as perfectly aligned image correspondences, but as approximate street-level matches.

5. Application: A Prototype Lighting-Aware Route Planner

5.1. Route Planning Implementation

To demonstrate a downstream application, we build a route-planning prototype that uses NightWalk’s per-image brightness predictions as edge weights in a pedestrian path planner over Greenwich Village. We construct a walking graph over the area from OpenStreetMap data and aggregate predictions spatially to each street block. Path planning uses the predicted lighting score as a multiplicative edge cost:

$$\text{cost}(e) = \ell(e) \cdot (1 + \lambda(1 - s_b(e))), \quad (2)$$

where $\ell(e)$ is edge length in meters and $\lambda \geq 0$ trades distance for lighting ($\lambda = 0$ recovers the shortest path). The recommended well-lit route between two endpoints is the shortest-cost path under $\text{cost}(\cdot)$, allowing pedestrians to prefer better-lit streets at the cost of a modest detour.

5.2. Qualitative Route Visualization

Figure 7 shows a representative example: a pedestrian walking ≈ 800 m from Warren Weaver Hall to the Waverly Diner. The shortest path (blue, 778 m) has a mean lighting score of 0.42. The well-lit path (pink, 1,057 m) routes through brighter blocks, achieving a mean lighting score of 0.74 at the cost of a 36% detour. This route was computed using the pessimistic metric p_b , which minimizes the worst-lit block along the path rather than optimizing average brightness. This example is a qualitative demonstration; brightness predictions carry noise from camera framing and exposure variation and should not be interpreted as a quantitative safety guarantee.

6. Limitations and Future Work

Our current implementation has several limitations that point to directions for future work.

Data collection. Our day–night pairing process is constrained by the scarcity of nighttime SVI, limiting geographic scope to a small area of New York City (Figure 8a).

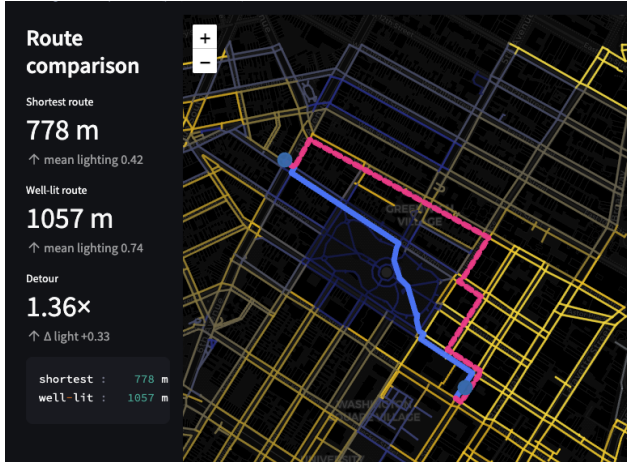


Figure 7. Lighting-aware vs. shortest pedestrian routes from Warren Weaver Hall to the Waverly Diner. **Blue**: shortest path, 778 m, mean lighting 0.42. **Pink**: well-lit path, 1057 m, mean lighting 0.74. Blocks are colored by predicted lighting score and gray blocks have no coverage.

Moreover, we had a very small set of nighttime images to fine-tune on since we had to collect them ourselves. Ideally, images would be collected in both conditions simultaneously or via a standardized protocol, reducing pairing inconsistencies and manual effort. Future work could also incorporate photometric calibration to ensure nighttime brightness more faithfully reflects luminance as perceived by a pedestrian, rather than exposure-adjusted pixel values.

Image confounders. Daytime images frequently contain parked cars, pedestrians, and other transient occlusions that affect apparent brightness independently of fixed lighting infrastructure. Future work could explore image processing techniques to filter or discount these confounding factors before extracting brightness-relevant features.

Feature detection. DINO-derived streetlight counts were substantially undersampled, as lampposts are frequently cropped out by the image frame. Panoramic imagery could address this by capturing the full scene at each location, and further prompt engineering could improve detection recall. Our current feature set and prompts were selected through manual inspection on a random subset of 100 images; a more systematic search over lighting-relevant features and Grounding DINO prompts may yield stronger pretraining signal.

Pretraining objectives. Our backbone ablation shows that supervised pretraining on object counts does not improve over ImageNet initialization for brightness prediction. Future work could explore pretraining targets more directly linked to illumination, such as sky fraction, surface reflectance, or time-of-day estimation from image timestamps.

7. Conclusion

We presented a pipeline for predicting nighttime street brightness from daytime SVI, addressing the challenge of limited nighttime imagery for urban lighting assessment. Our approach combines a DINO-based feature detection stage with an EfficientNet-B0 regression model fine-tuned on 974 manually paired day–night images. Our best model achieves $R^2 = 0.460$ on the held-out test set (using a pure ImageNet backbone), indicating that nearly half the variation in nighttime street brightness is predictable from daytime visual appearance alone.

Contrary to our hypothesis, task-specific pretraining on DINO-derived count targets did not improve over standard ImageNet initialization. Warmup scheduling and bounding-box supervision narrowed the gap only marginally, suggesting that at this dataset scale and noise level, count-based proxy supervision does not provide a stronger initialization than general ImageNet features. With a larger and more precisely paired dataset, task-specific pretraining may yet prove beneficial.

Qualitative analysis confirms that our brightness predictions are consistent with observed nighttime street appearance, and we demonstrate a downstream application to pedestrian route planning that successfully optimizes paths for nighttime illumination.

References

- [1] Joseph Breda, Daniel Campos Zamora, Shwetak Patel, and Jon E Froehlich. Nightlight: Passively mapping nighttime sidewalk light data for improved pedestrian routing. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, pages 1–13, 2025. 1
- [2] Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. A simple framework for contrastive learning of visual representations, 2020. 4
- [3] Zicheng Fan and Filip Biljecki. Nighttime street view imagery: A new perspective for sensing urban lighting landscape. *Sustainable Cities and Society*, 116:105862, 2024. 1, 2
- [4] Shilong Liu, Zhaoyang Zeng, Tianhe Ren, Feng Li, Hao Zhang, Jie Yang, Qing Jiang, Chunyuan Li, Jianwei Yang, Hang Su, Jun Zhu, and Lei Zhang. Grounding dino: Marrying dino with grounded pre-training for open-set object detection, 2024. 1, 2
- [5] Tingting Li Tianyi Ren Da Chen Wenjing Li Liu, Zhiyi and Waishan Qiu. Day-to-night street view image generation for 24-hour urban scene auditing using generative ai. *Journal of Imaging*, page 112, 2024. 2
- [6] Christopher Small, Christopher D. Elvidge, Deborah Balk, and Mark Montgomery. Spatial scaling of stable night lights. *Journal of Imaging*, pages 269–280, 2011. 1
- [7] Mingxing Tan and Quoc V. Le. Efficientnet: Rethinking model scaling for convolutional neural networks, 2020. 2
- [8] Brandon C. Welsh, David P. Farrington, and Stephen Douglas. The Impact and Policy Relevance of Street Light-

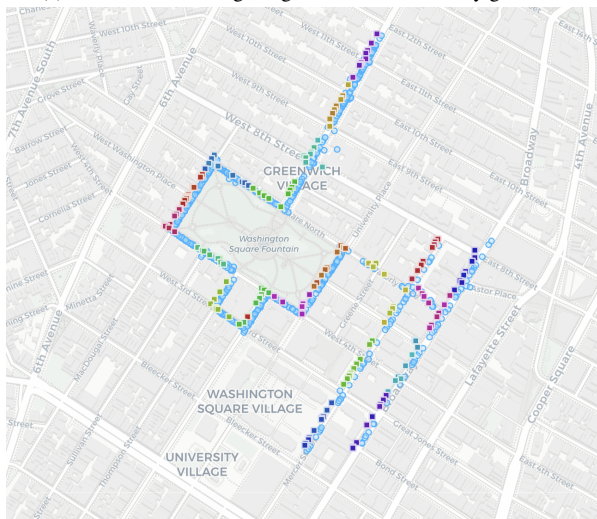
ing for Crime Prevention: A Systematic Review Based on
a Half-Century of Evaluation Research. *CrimRxiv*, 2024.
<https://www.crimrxiv.com/pub/wl9zqxga>. 1

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Supplementary Material



(a) EfficientNet training image locations, colored by grid cell.



(b) Train (blue) and test (colored) split locations. Test images evenly sample the collected routes.

Figure 8. Spatial distribution of images in downtown Manhattan (Greenwich Village / Washington Square area). Left: the full EfficientNet training set provides dense coverage across 321 grid cells. Right: the day-image train/test split, showing the test set is geographically uniform across all collected routes rather than clustered.

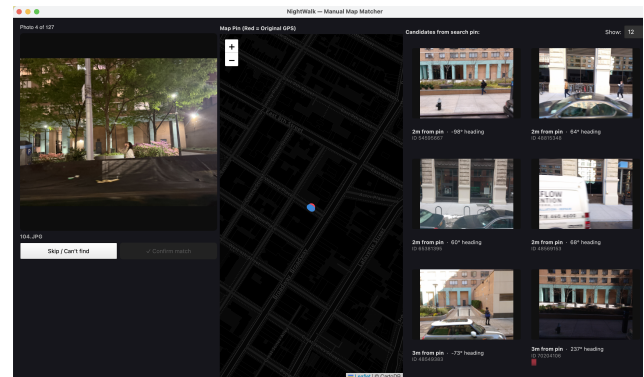


Figure 9. Manual map matcher interface for day–night image pairing. The annotator views a nighttime photograph (left) alongside an interactive map (center) and a grid of candidate daytime street-view images retrieved from the nearest GPS location (right). The annotator selects the best-matching daytime view or skips if no suitable match exists.